

Compare Results

Old File:

USB_PD_R3_2 V1.0 2023-10_Part25.pdf

10 pages (622 KB)

15/04/2024 10:53:04

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Appendix E FRS.pdf

10 pages (1.02 MB)

28/10/2024 11:27:53

Total Changes

247

Text only comparison

Content

- 197 Replacements
- 25 Insertions
- 25 Deletions

Styling and Annotations

- 0 Styling
- 0 Annotations

Go to First Change (page 1)

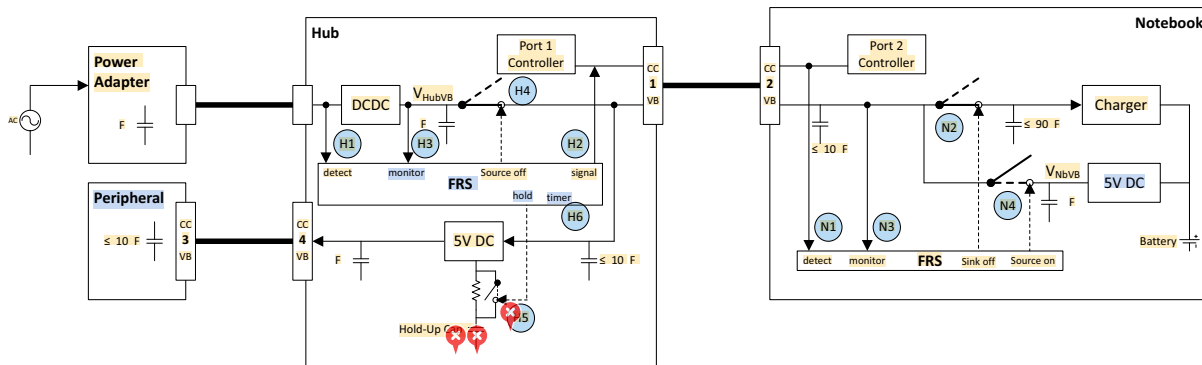
E FRS System Level Example

E.1 Overview

[Appendix E, "FRS System Level Example"](#) is intended to clarify *Fast Role Swap* (FRS) functionality at the system level through the use of an example implementation.

[Figure E.1, "Example FRS Capable System"](#) is an example of a *Hub* and *laptop* implementation that supports *Fast Role Swap* (see [Figure 7.16, "VBUS Power during Fast Role Swap"](#)). It is not the only possible *Hub* or *laptop* architecture. However, it is intended to provide an example system whose functionality is used here to illustrate how *Fast Role Swap* works.

Figure E.1 Example FRS Capable System



This appendix describes two cases that cover a variety of behaviors that might be seen in practice.

- Slow *VBUS* Discharge where *VBUS* between the *Hub* and the *laptop* takes more than 15ms (*t_{FRSwapInit}*) to discharge below 5.5 V (*vSafe5V* (max)). In this case the *FR_Swap* Message is sent by the *laptop* while *VBUS* is still greater than *vSafe5V* (max). See [Figure E.2, "Slow *VBUS* Discharge"](#).
- Fast *VBUS* Discharge where *VBUS* between the *Hub* and the *laptop* discharges very quickly, perhaps before the *Fast Role Swap Request* is even complete. See [Figure E.3, "Fast *VBUS* Discharge"](#).

However, neither the *Hub* nor the *laptop* can anticipate how quickly *VBUS* will discharge until the power adapter is disconnected from an *AC Supply* or it is unplugged from the *Hub*.

The *Fast Role Swap Request* is the momentary low driven by the *Hub* on the *CC* wire which is detected by the *laptop*.

[Figure E.2, "Slow *VBUS* Discharge"](#) and [Figure E.3, "Fast *VBUS* Discharge"](#) show the voltage seen on *VBUS* in relationship to the *Fast Role Swap Request*. They also show the transition between when the *Hub* stops supplying *VBUS* and when the *laptop* starts supplying *VBUS*.

Figure E.2 Slow VBUS Discharge

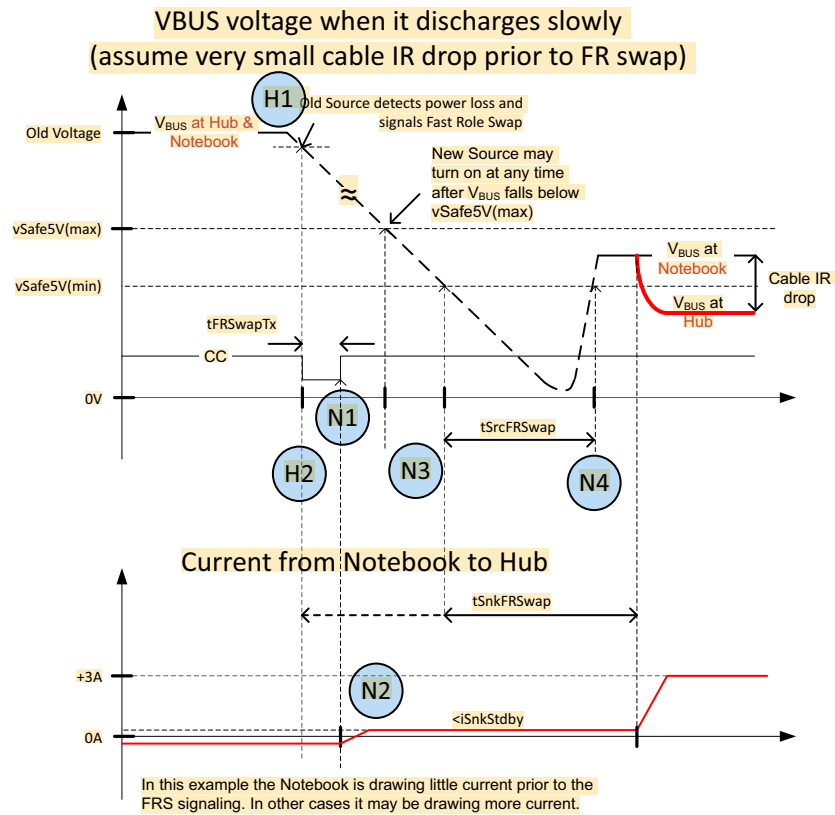
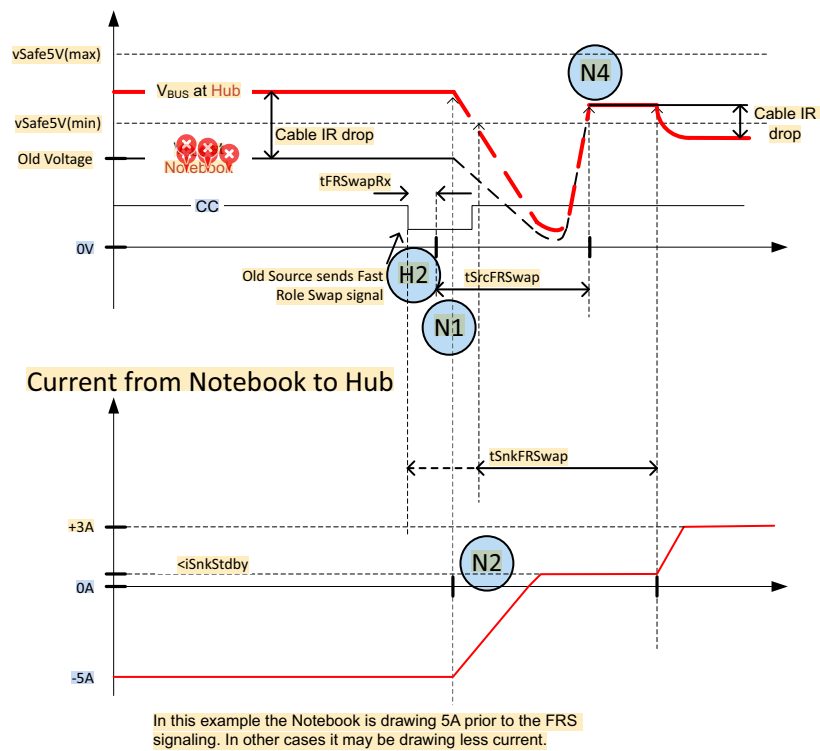


Figure E.3 Fast VBUS Discharge

VBUS voltage when it discharges quickly



E.2 FRS Initial Setup

Before a *Fast Role Swap* can occur, some initial setup steps are required. They require the laptop to discover whether *Fast Role Swap* is supported by the *Hub*, the amount of current the *Hub* requires after a *Fast Role Swap*, and whether the laptop is able and willing to provide that amount. They also ensure that the laptop supplies *VCONN* before, during and after an FRS. [Table E.1, "Sequence for setup of a Fast Role Swap \(Hub connected to Power Adapter first\)"](#) and [Table E.2, "Sequence for setup of a Fast Role Swap \(Hub connected to laptop before Power Adapter\)"](#) below show two typical sequences that might be used to prepare a laptop to support *Fast Role Swap*.

Table E.1 Sequence for setup of a Fast Role Swap (Hub connected to Power Adapter first)

Step #	Hub	Laptop
1	Hub connected to power adapter	
2	Hub is connected to laptop.	
3		Laptop sources 5 V to <i>VBUS</i> (<i>vSafe5V</i>). Laptop sources 5 V to <i>VCONN</i>
4		Laptop reads the cable to check its current carrying capability and/or if it is an <i>Active Cable</i> requiring <i>VCONN</i> .
5		Laptop sends a <i>Capabilities Message</i>
6	Hub sends a <i>Request Message</i>	
7	Hub and laptop establish an <i>Explicit Contract</i> with Hub as Sink.	
8		Laptop sends a <i>Get_Source_Cap</i> Message to determine how much power the Hub can provide.
9	Hub sends a <i>Source_Capabilities Message</i> with the <i>Dual-Role Power</i> bit set, and <i>Unconstrained Power</i> bit set, and <i>Maximum Current</i> > 0.	
10		Since the Hub can supply power the laptop sends a <i>PR_Swap Message</i>
11	Hub sends an <i>Accept Message</i> and starts supplying <i>VBUS</i>	
12		Laptop sends a <i>Get_Sink_Cap</i> Message to determine the current required by the Hub to support an FRS. If the Hub does not support FRS or the laptop cannot supply the required current, the laptop <i>Ignores</i> any <i>Fast Role Swap Requests</i> it might see.
13	If the Hub can supply more than 3A, it initiates a <i>VCONN Swap</i> to make to make itself the <i>VCONN Source</i> and reads the cable to check its current carrying capability.	
14	Hub sends a <i>Sink_Capabilities Message</i>	
15		Laptop sends a <i>Request Message</i>
16	Hub and laptop establish an <i>Explicit Contract</i> with Hub as source.	

Table E.1 Sequence for setup of a Fast Role Swap (Hub connected to Power Adapter first)

Step #	Hub	Laptop
17		If the laptop has detected that it is connected via an Active Cable (or one that supports Alternate Modes) and/or that it can support an FRS , it initiates a VCONN Swap to make itself the VCONN Source . This removes a requirement that the Hub to hold up VCONN during the FRS .
18	Normal PD Power traffic flow	
19	The Hub and laptop are now ready to do a Fast Role Swap in case the power adapter gets removed.	

Table E.2 Sequence for setup of a Fast Role Swap (Hub connected to **laptop before **Power Adapter**)**

Step #	Hub	Laptop
0	Hub is connected to laptop .	
1		Laptop sources 5 V to V_{BUS} (vSafe5V). Laptop sources 5 V to VCONN
2		Laptop reads the cable to check its current carrying capability and/or if it is an Active Cable requiring VCONN .
3		Laptop sends Source_Capabilities Message
4	Hub sends Request Message	
5	Hub and laptop establish an Explicit Contract with Hub as Sink .	
6		Laptop sends a Get_Source_Cap Message to determine how much power the Hub can provide
7	Hub sends a Source_Capabilities Message with the Dual-Role Power bit set, and Unconstrained Power bit cleared, and Maximum Current = 0.	
8		Since the Hub cannot supply power, the laptop does not send a PR_Swap Message
9	The power adapter is connected to the Hub	
10	If the Hub can source more than 3A, it initiates a VCONN Swap to become the VCONN Source .	
11	Hub reads the e-marker to determine the cable's current carrying capability.	
12	Hub initiates a Power Role Swap to become the Source	
13	Hub sends a Source_Capabilities Message with the Unconstrained Power bit set and Maximum Current > 0.	
14	Hub and laptop establish an Explicit Contract with Hub as source.	
15		Laptop sends a Get_Sink_Cap Message to determine the current required by the Hub to support an FRS . If the Hub does not support FRS or the laptop cannot supply the required current, the laptop Ignores any Fast Role Swap Requests it might see.

Table E.2 Sequence for setup of a Fast Role Swap (Hub connected to laptop before Power Adapter)

Step #	Hub	Laptop
16		If the laptop has detected that it is connected via an Active Cable (or one that supports Alternate Modes) and/or that it can support an FRS , it initiates a VCONN Swap to make itself the VCONN Source . This removes a requirement that the Hub also hold up VCONN during the FRS .
17	The Hub and laptop are now ready to do a Fast Role Swap in case the power adapter gets removed.	

E.3 FRS Process

After the initial setup is completed and the laptop has determined both that the Hub can request FRS and that the laptop is able and willing to supply the requested current, the system is ready to support FRS. This section describes the sequence of events that take place during a Fast Role Swap. The following figures and tables do not cover the actions of the Device Policy Manager or the Policy Engine. Those actions occur orthogonally to the electrical events shown in this appendix. However, the diagrams do indicate the inputs/outputs where the DPM and Policy Engine interact with the electrical events:

- The laptop sends the **FR_Swap** Message to initiate the FRS AMS (see [Figure 7.43, "Transition Diagram for Fast Role Swap"](#)) within 15ms after the laptop detects the Fast Role Swap Request on CC.
- The laptop sends the final **PS_RDY** Message in the FRS AMS only after it is sourcing V_{BUS} .

Figure E.4 Slow V_{BUS} discharge after FR_Swap message is sent

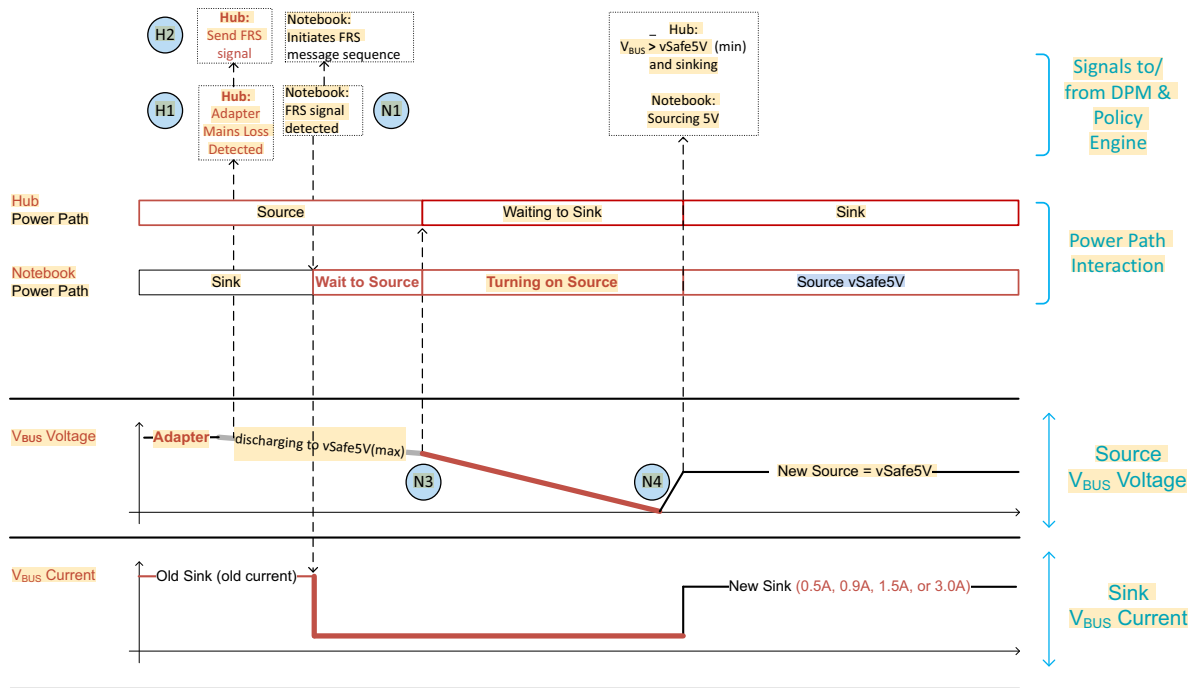


Table E.3 Sequence for slow VBUS discharge (it discharges after FR_Swap message is sent)

Step #	Hub	Laptop
1	The power adapter's AC Supply power is lost.	
2	Hub detects the power adapter disconnect (H1) as quickly as possible.	
3	Hub sends <i>Fast Role Swap Request</i> on CC (H2) and starts monitoring V_{HubVB} (H3). Hub also starts a <i>tSnkFRSwap</i> timer after the FRS signal begins and <i>VBUS</i> has fallen below <i>vSafe5V</i> (min).	
4		Laptop detects <i>Fast Role Swap Request</i> on CC (N1) that triggers sending of the <i>FR_Swap Message</i> . This can happen at any point in the following steps so long as it is within 15 ms (<i>tFRSwapInit</i>).
5		Laptop opens the sinking switch (N2), as quickly as possible to minimize power drained from Hub after the <i>Fast Role Swap Request</i> .
6		Laptop begins monitoring <i>VBUS</i> (N3) to know when to turn the laptop into a Source.
7	Hub opens the sourcing switch (H4) while $V_{HubVB} > 5.5V$ (after the <i>Fast Role Swap Request</i> is sent). However, the sourcing switch (H4) must be kept closed until V_{HubVB} is as close to 5.5V as possible. It is important for the Hub to open its sourcing switch (H4) before the laptop's sourcing switch (N4) gets closed to minimize inrush current.	
8	Hub closes the switch (H5) to use the hold-up capacitor to supply <i>VBUS</i> to the peripheral(s). Systems with a holding cap permanently in place do not need the switch (H5). Hub does not draw more than <i>iSnkStdby</i> from <i>VBUS</i> , until the <i>tSnkFRSwap</i> timer expires.	
9		Laptop detects $VBUS < V_{NbVB}$ (N1) before closing the sourcing switch (N4) when V_{NbVB} is as close as possible to 5.5V. This minimizes the time when <i>VBUS</i> is not sourced.
10		Laptop closes sourcing switch (N4). When this occurs the Hub's input capacitance on <i>VBUS</i> will be less than $10\mu F$ (<i>cSnkBulk</i>).
11	Hub's <i>tSnkFRSwap</i> timer expires (H6).	
12	Hub draws up to the current it <i>Advertised</i> in the <i>Fast Role Swap required USB Type-C Current</i> field of its <i>Sink_Capabilities Message</i> .	
13	Hubs with (H5) will open (H5) and remove the Hold-Up capacitor.	

Figure E.5 VBUS discharges quickly before FR_Swap message is sent after adapter disconnected

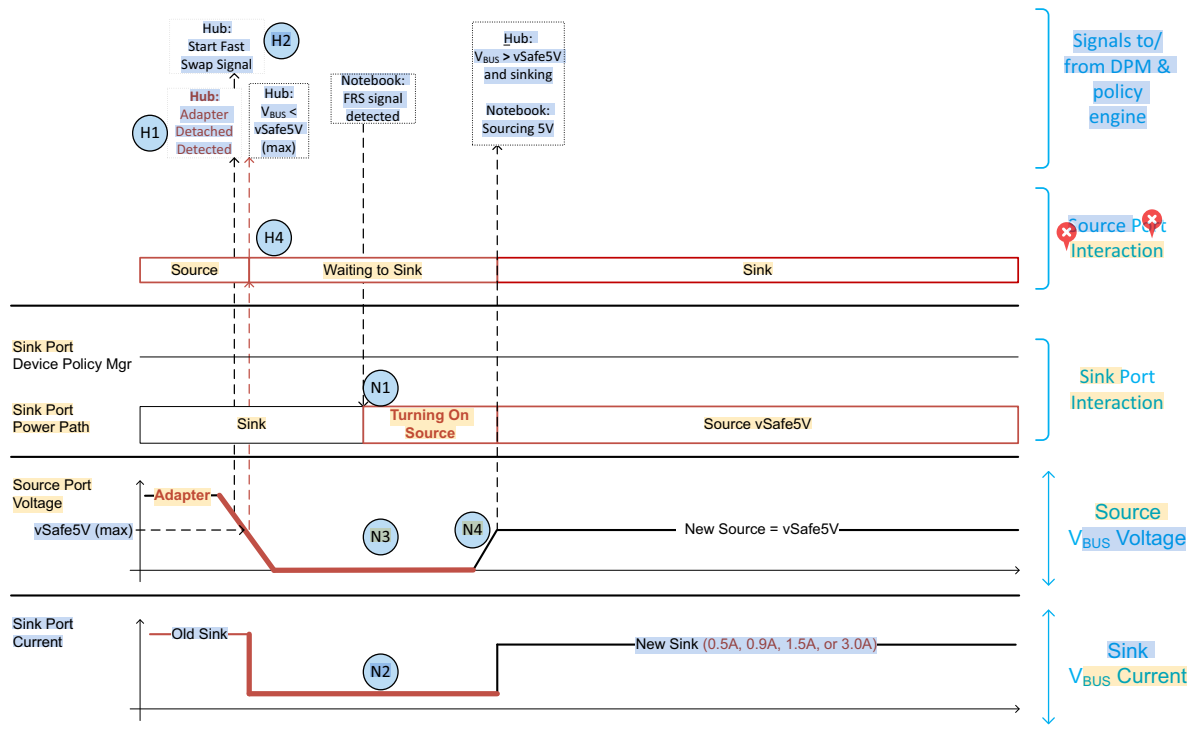


Table E.4 VBUS discharges quickly after adapter disconnected

Step #	Hub	Laptop
1	The power adapter is Detached from the Hub.	
2	Hub detects power adapter disconnect (H1)-causing V_{HubVB} to drop below 5.5V very rapidly.	
3	Hub sends Fast Role Swap Request on CC (H2) and starts monitoring V_{HubVB} (H3). Hub opens sourcing switch (H4). Hub also starts a $t_{SnkFRSwap}$ timer.	
4	Hub closes the switch (H5) to use the hold-up capacitor to supply V_{BUS} to the peripheral(s). Systems with a holding cap permanently in place do not need the switch (H5). Hub does not draw more than $i_{SnkStdby}$ from V_{BUS} , until the $t_{SnkFRSwap}$ timer expires.	
5		Laptop detects Fast Role Swap Request on CC (N1) that triggers sending of the FR_Swap Message . This can happen at any point in the following steps so long as it is within 15 ms ($t_{FRSwapInit}$).
6		Laptop opens the sinking switch (N2), as quickly as possible to minimize power drained from Hub after the Fast Role Swap Request .
7		Laptop begin monitoring V_{BUS} (N3) to know when to turn the laptop into a Source

Table E.4 VBUS discharges quickly after adapter disconnected

Step #	Hub	Laptop
8		Laptop detects $V_{BUS} < V_{NbVB}$ (N3).
9		Laptop closes sourcing switch (N4). When this occurs the Hub's input capacitance on V_{BUS} will be less than 10 μF ($c_{SnkBulk}$).
10	Hub's $t_{SnkFRSwap}$ timer expires (H6).	
11	Hub draws up to the current it <i>Advertised</i> in the <i>Fast Role Swap required USB Type-C Current</i> field of its <i>Sink_Capabilities</i> Message.	
12	Hubs with (H5) will open (H5) and remove the Hold-Up capacitor.	